MTH 107(10) Dr. James Ashe

Final Exam Review The Final is on Wednesday 3$^{\text{rd}}$ December, 2014 from 2:30pm-4:30pm

1. Using the symbolic representations
b: I ride a bicycle to school.
s: It is snowing.
r: It is raining.
express the following statements in symbolic form.

a. It is raining and I rode my bicycle to school.

$r \wedge b$

b. If it is raining or snowing, then I will not ride my bicycle to school.

$(r \vee s) \to \sim b$

2. Using the statement, "If your order is over \$50, then you will not have to pay shipping fees." and parts (a), (b), and (c) below, complete the following:

a. If you do not have to pay shipping fees then your order is over \$50.
b. If your order is not over \$50 then you have to pay shipping fees.
c. If you have to pay shipping fees then your order is not over \$50.

Identify the converse of the statement. ______
Identify the inverse of the statement. ______
Identify the contrapositive of the statement. ______

3. For parts 3a-3d, identify the negation of the provided statement.

a. Some students are taking Math 107.
(a) All students are taking Math 107.
(b) No students are taking Math 107.
(c) Some students are not taking Math 107.
(d) Some students are taking Math 107.

b. All players need to attend practice.
(a) All players need to attend practice.
(b) No players need to attend practice.
(c) Some players do not need to attend practice.
(d) Some players need to attend practice.

c. No professors know how to read.
(a) All professors know how to read.
(b) No professors know how to read.
(c) Some professors do not know how to read.
(d) Some professors know how to read.

d. Some students are going to lunch.
(a) All students are going to lunch.
(b) No students are going to lunch.
(c) Some students are not going to lunch.
(d) Some students are going to lunch.

4. Use a truth table to determine if the following arguments are valid.

a. b: You buy the bike. t: You buy the tent.

1. You buy the bike or the tent.
2. You buy the bike.

Therefore, you do not buy the tent.

		$\boxed{1}$	$\boxed{2}$		$\boxed{c}$	$\boxed{\text{Argument}}$
b	t	$b \vee t$	b	$1 \wedge 2$	$\sim t$	$(1 \wedge 2) \to c$
T	T					
T	F					
F	T					
F	F					

b. r: You register early. d: You receive a discount.

1. If you register early, then you will receive a discount.
2. You register early.

Therefore, you will receive a discount.

		$\boxed{1}$	$\boxed{2}$		$\boxed{c}$	$\boxed{\text{Argument}}$
r	d	$r \to d$	r	$1 \wedge 2$	r	$(1 \wedge 2) \to c$
T	T					
T	F					
F	T					
F	F					

5. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ be the universal set, and $A = \{1, 2, 3, 4\}$ and $B = \{4, 6, 8\}$ be subsets of U. Find the indicated set, and draw a Venn diagram corresponding to the indicated set.

a. $A \cup B$
b. $A \cap B$
c. A'
d. B'
e. $A \cap B'$
f. $A' \cup B$
g. $(A \cap B)'$
h. $(A \cup B)'$

6. A college has 1105 students. This semester 320 students are taking a math course, 405 of the students are taking a history class, and 250 of the students taking a math class are not enrolled in a history class.

a. Construct a Venn diagram, and write the appropriate numbers in the four regions
b. How many of the students are taking math and history?
c. How many students are taking history but not math?
d. How many students are taking math or history?
e. How many students are taking neither math nor history?

7. *Survivor* and *CSI* belong to the Thursday night prime time lineup. A group of 600 college students were surveyed. 350 students watch *Survivor* and 280 watch *CSI*. 180 students watch both shows.

a. Construct a Venn diagram, and write the appropriate numbers in the eight regions.
b. How many students watch only *Survivor*?
c. How many students watch *CSI* or *Survivor*?
d. How many students watch only *CSI*?

8. Find the value of each of the following:

a. $4!$
b. $7!$
c. $0!$
d. $\dfrac{10!}{7!}$
e. $\dfrac{27!}{4!25!}$
f. $\dfrac{1000!}{998!}$
g. ${}_7P_2$
h. ${}_6C_4$

9. A certain model of pickup truck is available in five exterior colors, three interior colors, and three interior styles. In addition, the transmission can be either manual or automatic, and the truck can have either two-wheel or four-wheel drive. How many different versions of the pickup truck can be ordered?

10. A certain fraternity must elect a president, vice president, and treasurer from its 30 members. How many ways can this be done?

11. A three-person committee must be chosen from a group of ten men and twelve women. How many committees are possible if it must consist of the following?

a. Three men.
b. Three women.
c. One woman and two men.
d. One man and two women.
e. One man and two women, and one of these is selected as a spokesperson of the committee.

12. A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following.

a. A red card
b. A queen
c. A club
d. The queen of clubs
e. A queen or a club
f. not a queen

13. Three coins are tossed. Find each of the following:

a. the sample space
b. the event, E_1, that exactly two are tails.
c. the event, E_2, that at least two are tails.
d. the probability of E_1.
e. the probability E_2.

14. A drawer has 7 socks. 4 socks are black and 3 are white socks. John randomly pulls out 4 socks. Find the probability of the following:

a. all 4 socks are black.
b. exactly 2 are white.
c. at least 3 are white.

15. Of all the flashlights is a large shipment, 15% have a defective bulb, 10% have a defective battery, and 5% have both defects. If you buy one of these flashlights, find the probability of the following:

a. a defective bulb or a defective battery
b. a good bulb or a good battery
c. a defective bulb and a defective battery.
d. just a defective bulb

16. Use the provided table to answer the following questions.

Age	**Obama**	**McCain**	**Other/NA**
18-29	11.9%	5.8%	0.4%
30-44	15.1%	13.3%	0.6%
45-64	18.5%	18.1%	0.4%
65+	7.2%	8.5%	0.3%

a. What is the probability that a voter under 45 voted for McCain.

b. What is the probability that a person who voted for McCain is 45 or older.

17. Gregor's Garden Corner buys 40% of its plants from the Green Growery and the rest from Herb's Herbs. Of the plants from the Green Growery, 20% must be returned, and 10% of those from Herb's Herbs must be returned.

a. draw a tree diagram describing the probability
b. find the probability that a plant must be returned and it is from Herb's Herbs.
c. find the probability that a plant from the Green Growery must be returned.
d. find the probability that a plant must be returned.

18. Sixteen randomly selected families were surveyed to determine the number of children in each family. The following results were found:

3.7	3.6	4.0	3.8	4.1	3.9	3.5	3.9
3.6	4.0	3.6	3.7	3.9	3.3	3.6	4.0

a. Find the mean.
b. Find the median.
c. Find the standard deviation.
d. Construct a histogram using 5 subintervals, beginning at 3.2, and ending at 4.2.

19. Find the following probabilities,

a. $p(z < 1.35)$

b. $p(0 < z < 2.02)$

c. $p(-.09 < z < 1.06)$

d. $p(-1.1 < z < -.08)$

20. A population is normally distributed with a mean 78 and standard deviation 7.5. Find the following probabilities.

a. $p(x < 70)$

b. $p(70 < x < 85)$

c. $p(80 < x < 95.5)$

d. $p(95.5 < x < 108)$

21. Find c such that each of the following is true.

a. $p(z < c) = 0.2776$

b. $p(z > c) = .0351$

c. Let $\mu = 100$ and $\sigma = 15$ find c such that $p(x < c) = 27.76\%$

22. The time it takes latex paint to dry is normally distributed with $\mu = 3.5$ hours and $\sigma = .75$ hours. Find the probability that drying time will be as follows.

a. less than 2.25 hours.

23. The average score on the ACT is 18 with a standard deviation of 6. If a college only wants to accept students with an ACT score at least in the top 25%. What is the minimum score they should accept? What about the top 5%?